BLUESTAQ™ | NDSU NORTH DAKOTA STATE UNIVERSITY

A good day to plant.

A day worth making count.

Dr. Mike Ortiz

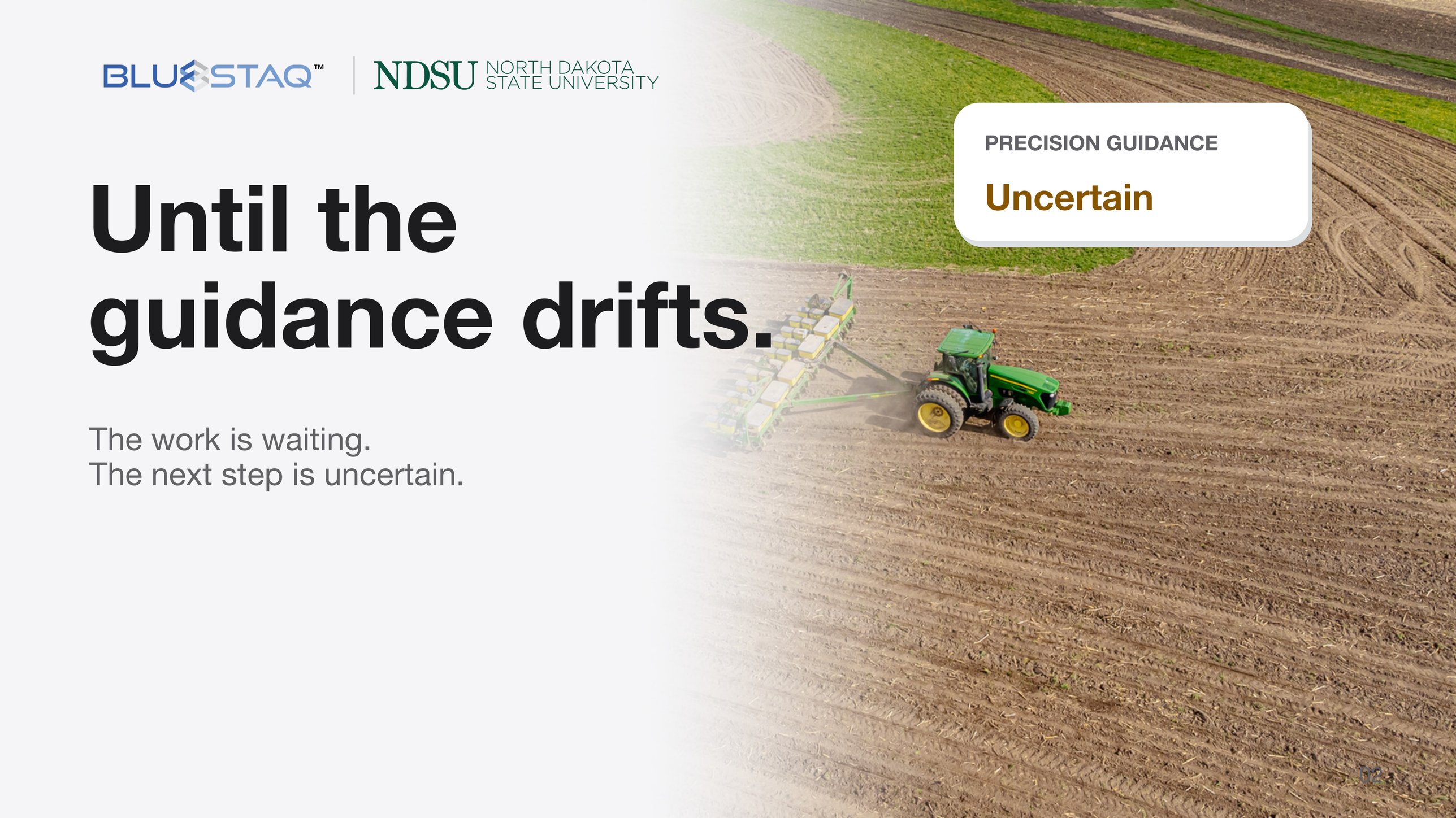
Principal Research Scientist
and Engineering Manager



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Until the guidance drifts.

The work is waiting.
The next step is uncertain.



What do you trust now?

The Sun?

Interference?

The machine?

Which evidence would change your mind?

The precision operation has paused.

Food security is national security.

One drifting tractor is one farm’s bad morning.
The dependency behind it reaches the whole nation.



Earth: NASA / Apollo 17 crew · December 7, 1972

WHY THIS MATTERS

Food security has four parts.

Available

Enough food is produced or brought in.

Disruption hits here

Accessible

People can afford it and reach it.

Usable

It is safe and nourishing.

Stable

All of this holds through shocks.

Disruption hits here

Precision agriculture protects availability and stability.

FAO · World Food Summit, 1996 · Food Security policy brief, 2006

WHY THIS MATTERS

Grain is strategy.

1972

A surprise purchase

The Soviet Union bought about 18 million tons of U.S. grain. USDA was caught off guard.

1974–77

Forecasting from orbit

NASA, USDA and NOAA used Landsat to estimate wheat in the U.S., Canada and the USSR.

2010 · 2022

Shocks travel fast

Drought and an export ban in Russia; then war disrupts Black Sea exports. Satellites track harvests no one can visit.

USDA ERS 1973 · GAO 1973 · NASA LACIE 1978 · USDA ERS 2011 · UN · NASA Harvest

WHY THIS MATTERS

The nation treats it as critical.

2004 · HSPD-9	Defend agriculture and food against attacks, disasters and emergencies.
Critical infrastructure	Food and Agriculture is one of 16 U.S. critical infrastructure sectors.
2020 · EO 13905	Make critical systems resilient to disruption of GPS and other PNT.
2025 · USDA	National Farm Security Action Plan.

“Farm security is national security.”

HSPD-9 · PPD-21 and NSM-22 · CISA · EO 13905 · Quote: USDA release headline, July 8, 2025

CLOSE TO HOME

North Dakota feeds the nation.

#1 in the U.S. in 2024 for:

- **Spring wheat**
- **Durum wheat**
- **Canola**
- **Sunflowers**
- **Dry edible beans**
- **Flaxseed**

The challenge fleet works near Casselton, in Cass County.

USDA NASS, North Dakota Agricultural Statistics 2025
Image: NASA JSC · ISS Expedition 37 · Sept. 15, 2013



WHY THIS MATTERS

The modern farm runs on space.

58%

of U.S. corn acres planted with auto-steer guidance

USDA ERS · 2016 survey
Up from 5% in 2001

$1.4T

in U.S. private-sector benefits from GPS

RTI for NIST · 1984–2017
2017 dollars

$1B

estimated loss per day in a GPS outage

Up to 50% higher during planting season

When guidance fails, the planting window does not wait.

USDA Economic Research Service · RTI International for NIST, 2019

One dependency. Many ways to break it.

Nature · May 2024

An extreme (G5) geomagnetic storm degraded GPS guidance mid-planting.

Interference · 2024

Eastern Finland farmers report jamming degrading tractor guidance.

Same symptom. Different cause. Different response.

Aurora: NASA / Bill Dunford · May 11, 2024

Four ways space feeds a nation.



Navigate

Straight rows, less overlap, less waste.

GPS and other GNSS
RTK corrections



Observe

Crop health and soil moisture, season after season.

Landsat · Sentinel-2
NASA SMAP



Forecast

Weather that decides when to plant and harvest.

NOAA GOES · JPSS



Warn

Solar-storm warning, about 15–60 minutes ahead.

NOAA SOLAR-1 at L1
NOAA SWPC

Space is part of the farm’s infrastructure.

NASA · NOAA NESDIS and SWPC · USGS · ESA · See “How space feeds a nation: sources”

An echo is a measurement.

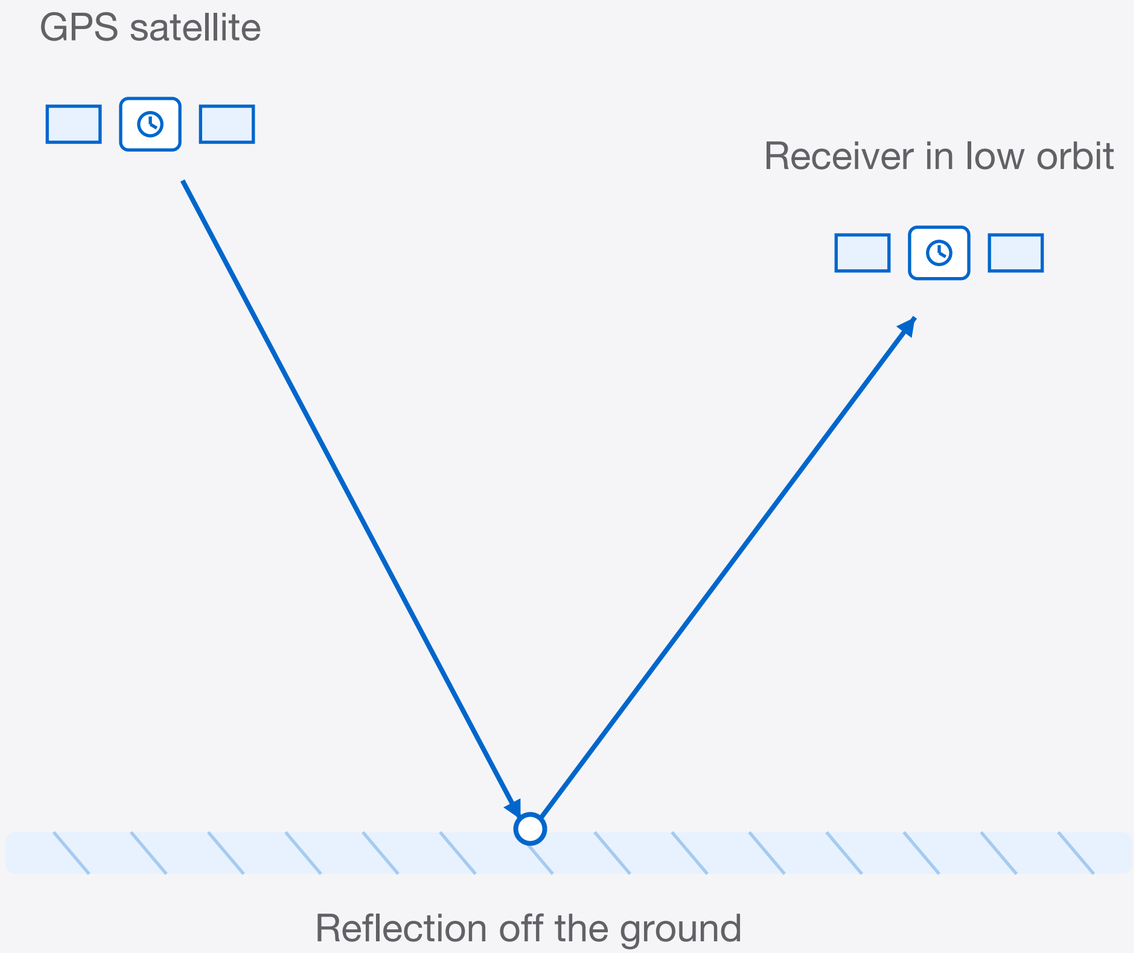
In the field

A brief reflection near a headland: dismiss the false alarm.

From orbit

NASA’s CYGNSS satellites turn reflected GPS signals into soil-moisture estimates.

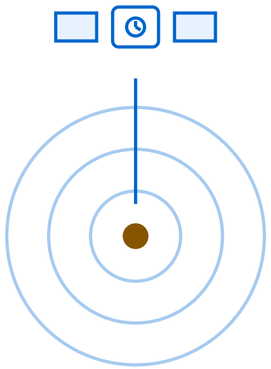
Same physics. Different question.



Conceptual geometry · NASA CYGNSS

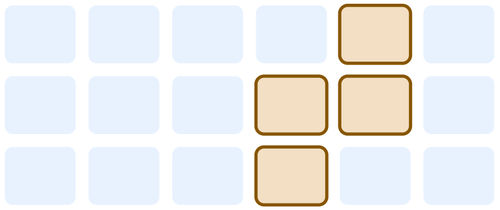
Listening for interference.

From orbit



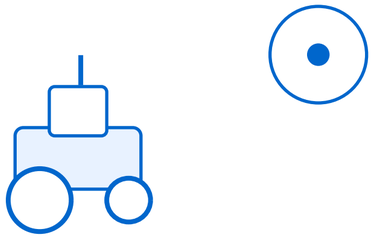
Commercial satellites detect and geolocate GPS jamming.

From the air



Aircraft accuracy reports map likely interference.

In the field

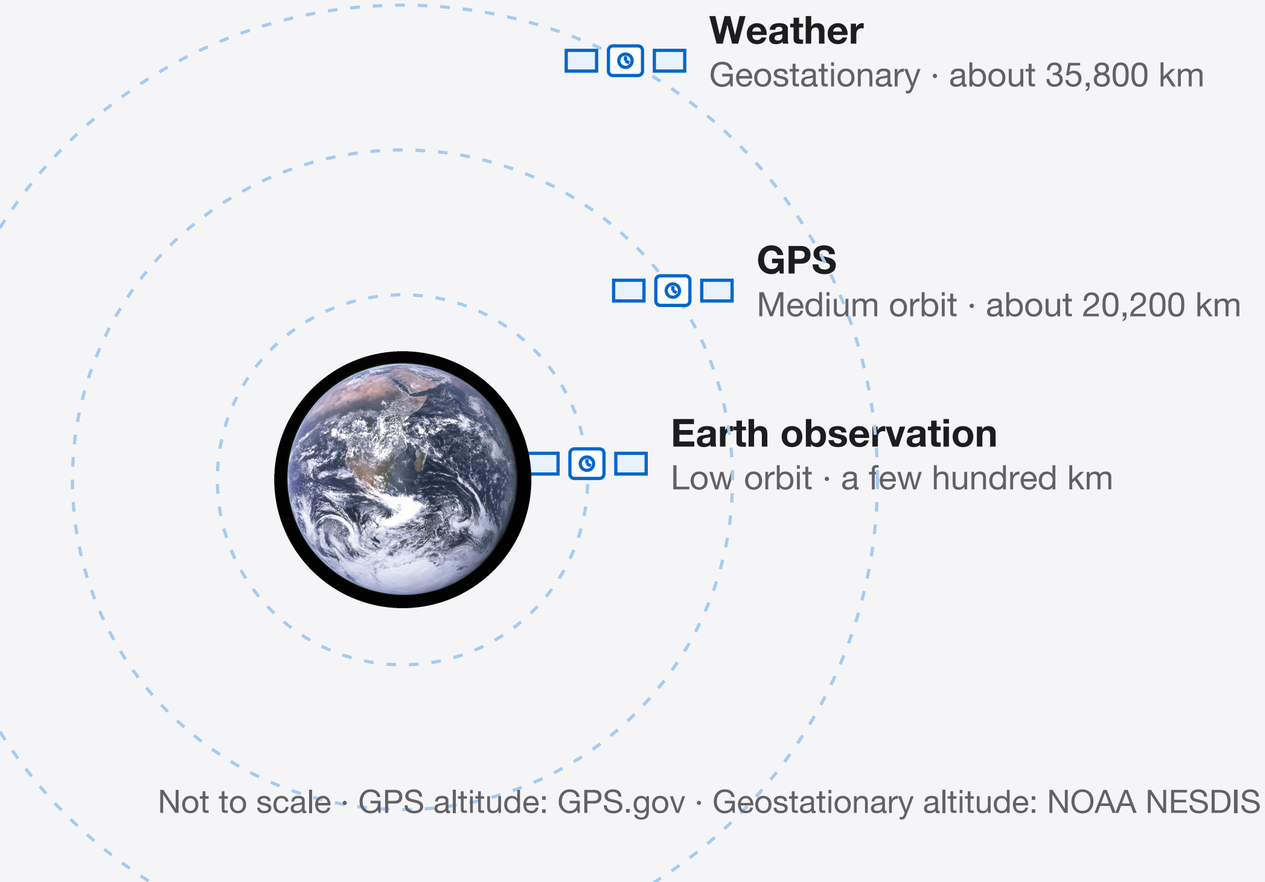


Your tractors, plus delayed and imperfect reports.

No single view is proof. Corroboration is the skill.

WHERE BLUESTAQ WORKS

Space Domain Awareness: watch the layer we depend on.



Not to scale · GPS altitude: GPS.gov · Geostationary altitude: NOAA NESDIS

What is up there?

Satellites, debris, and where they are going.

Is it working?

Health and status of the satellites we rely on.

What is space doing?

Solar storms, the ionosphere, and interference.

Your challenge lives in the third question.

Meet Bluestaq.

Defense-grade data infrastructure, proven across national security missions, now for every organization.

2018

Founded in Colorado Springs

First built for the Space Force's Space Domain Awareness mission

300+

data types in the Unified Data Library

Bluestaq is its primary developer

200M+

records pulled from the UDL every day

Up to 70 applications 2024 report to Congress

"The question is the same on a farm as on a mission."

Bluestaq About page · Space Systems Command, Mar. 2024 · Breaking Defense, June 2024 · Quote: Bluestaq /ARQ for agriculture, 2026

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Your speakers and judges.

MO

Dr. Mike Ortiz

Principal Research Scientist
and Engineering Manager

Host · Opening remarks

SH

Dr. Seth Harvey

Co-Founder and
Chief Executive Officer

Speaker · Judge

SS

Shane Sherman

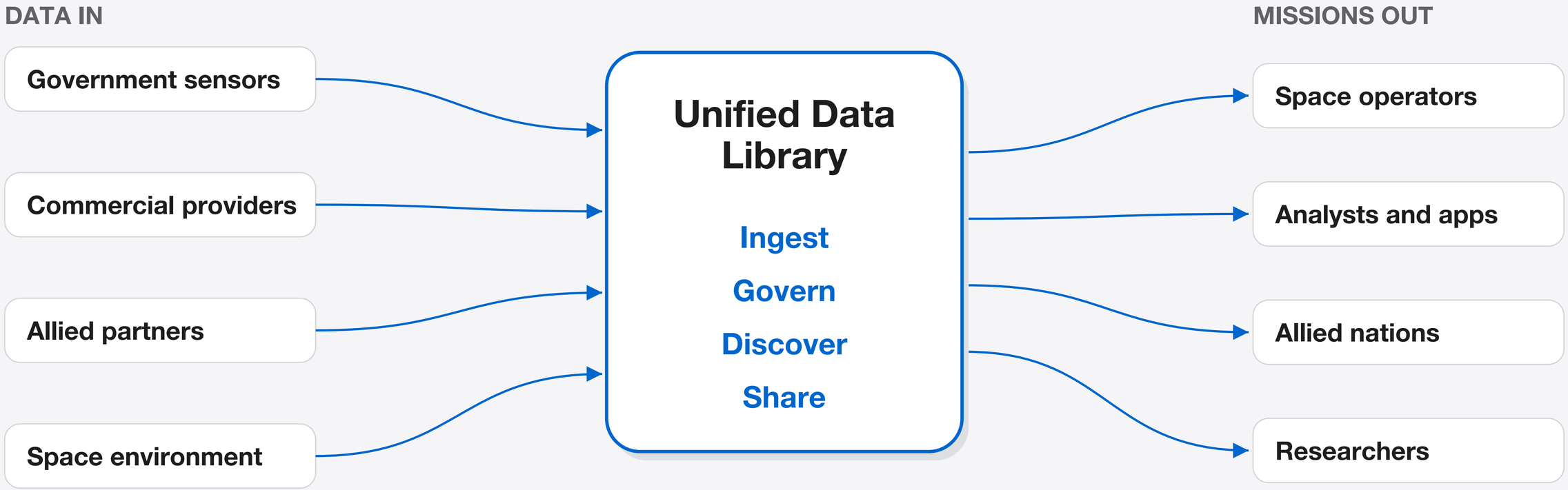
Chief Financial Officer

Speaker · Judge

Judges score every demonstration with the published 100-point rubric.

THE UNIFIED DATA LIBRARY

One library. Many sources. Many missions.



Commercial data in the UDL helped locate a vessel suspected of illegal fishing off Peru.

Space Systems Command, Mar. 11, 2024 · Breaking Defense, June 27, 2024 · Bluestaq · Simplified; not every source or user is shown

FROM SIMULATION TO MISSION

Your challenge, in real-world data.

IN YOUR CHALLENGE		IN THE REAL WORLD
sgi_indices.json Kp and Dst geomagnetic indices	→	**Sgi** Space-weather indices, including Kp and Dst
scintillation_series.json Regional S4 scintillation proxy	→	**IonoObservations** Ionospheric observations · S4 is simplified here
rfi_reports.json Delayed, uncertain interference reports	→	**Emireport** Spectrum-interference reports, including GPS
constellation_advisories.json Notices about satellite status	→	**Notification** Community notices and alerts
tractor_telemetry.csv.gz Satellites, C/N0, fix state, position	→	**GnssObservationset** GNSS receiver observations and status

Different owners, clocks and reliability. Joining them is the job.

Data-type names from Bluestaq’s public UDL SDK (github.com/Bluestaq, Apache-2.0) · All challenge data is simulated

Someone has to decide.

You can help.

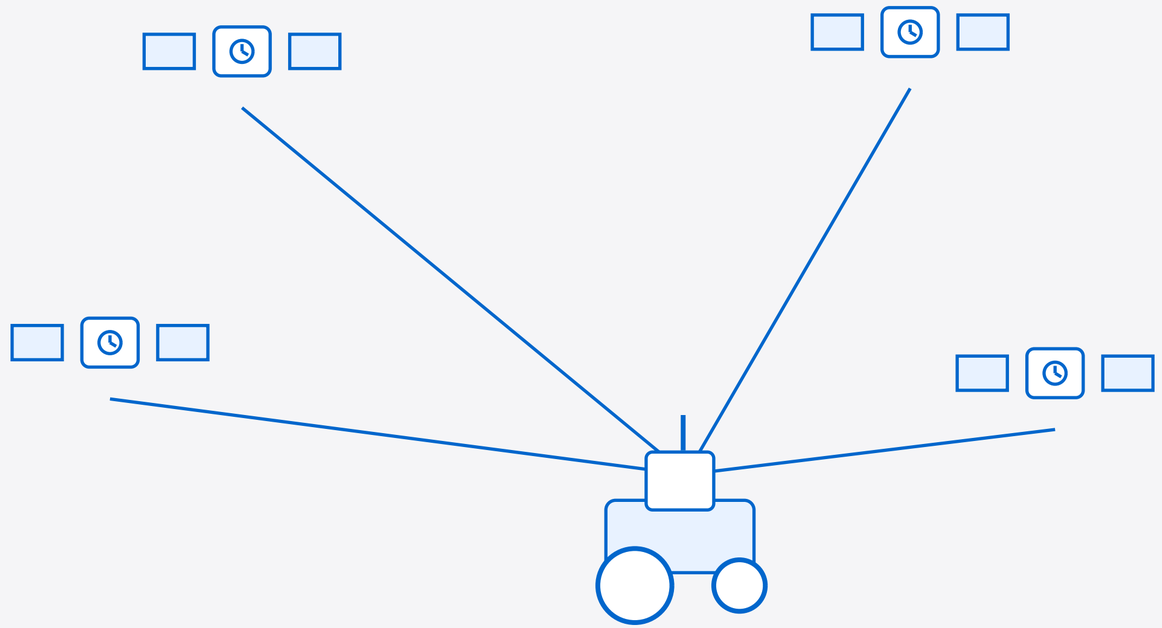
A better call in one cab,
repeated across a nation’s farms.

Navigation is a timing problem.

Travel time × speed of light = apparent distance

1 microsecond

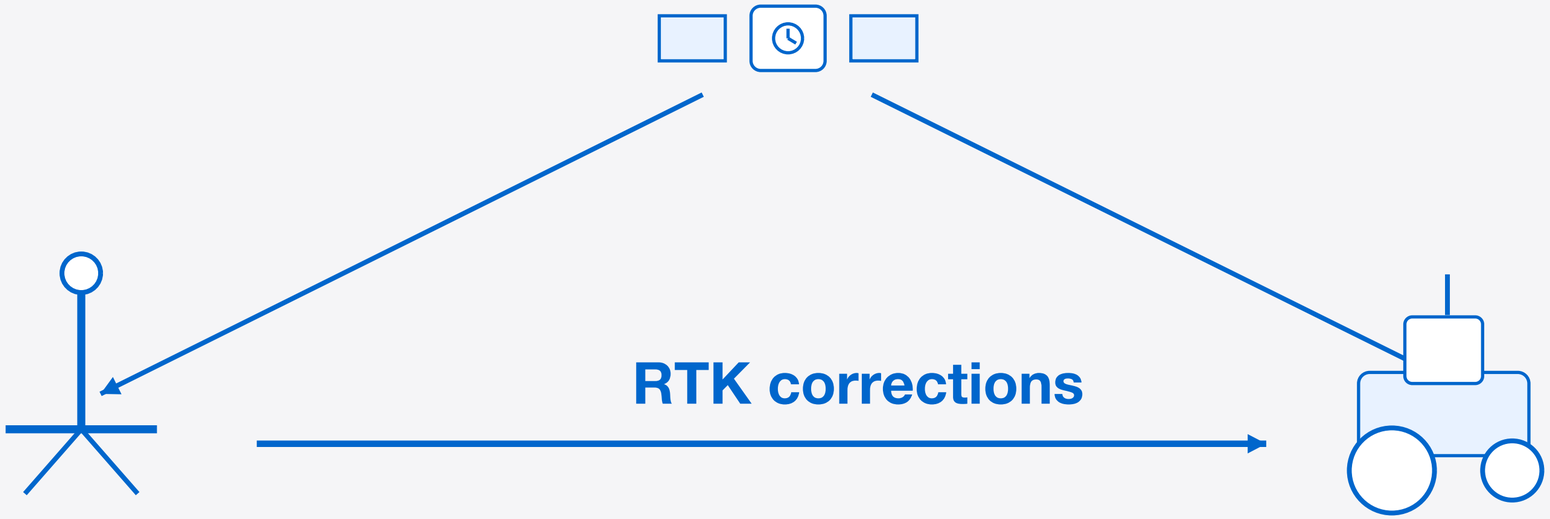
About 300 meters of range.



Position + receiver clock error

GNSS: satellite navigation · GPS is one constellation

Precision needs a second reference.



Base station

Known position

Moving receiver

Position relative to base

Precision depends on usable satellite signals and a correction link.

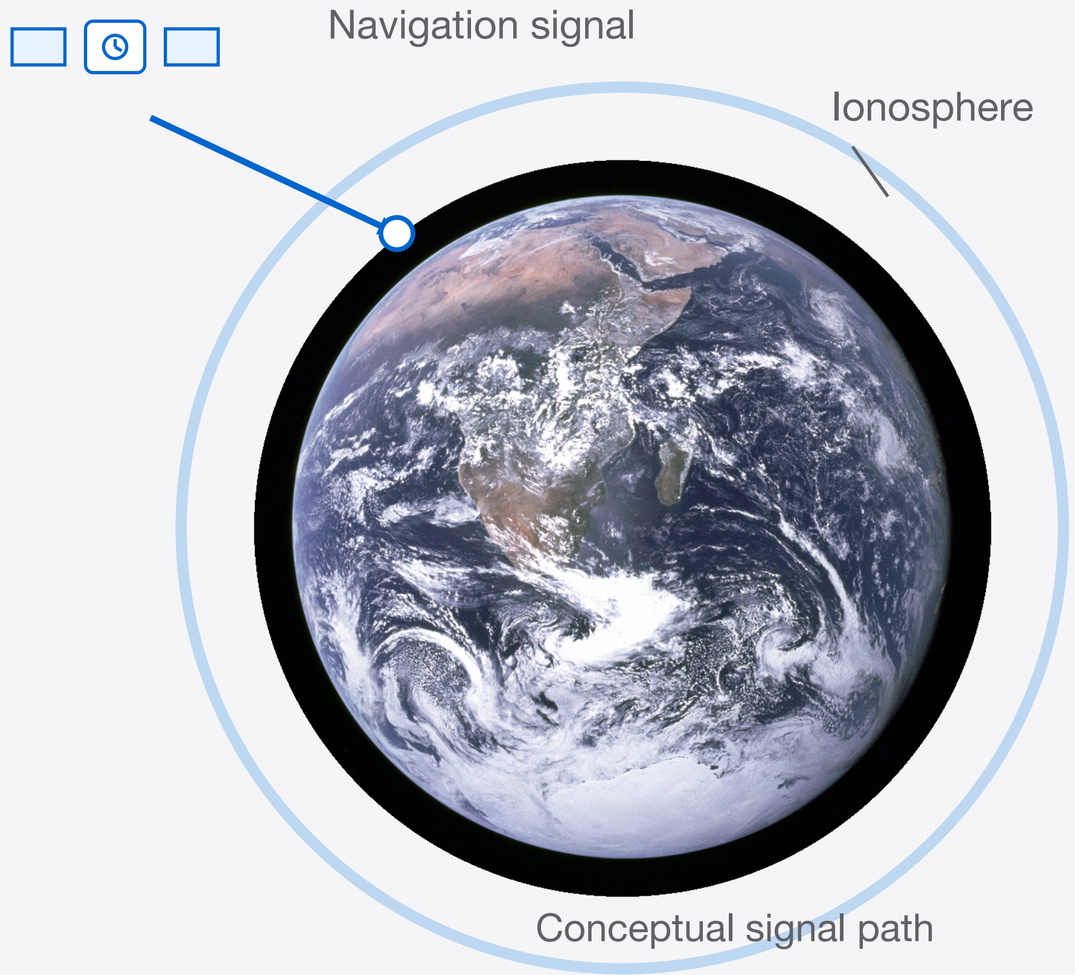
Space has weather, too.

Changed delay.

Signal flicker.

Tracking can fail.

Think of light shimmering
through moving water.

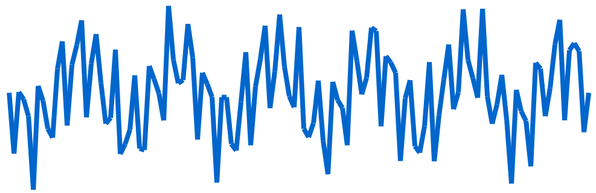


Radio propagation · Simplified geometry

Noise. Deception. Echoes.

Jamming

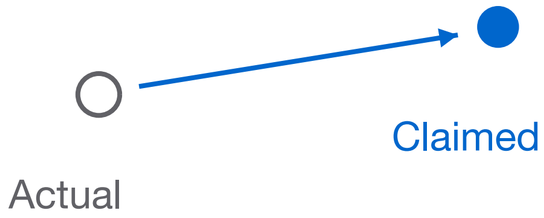
Authentic signal obscured



A quiet voice in a noisy room.

Spoofing

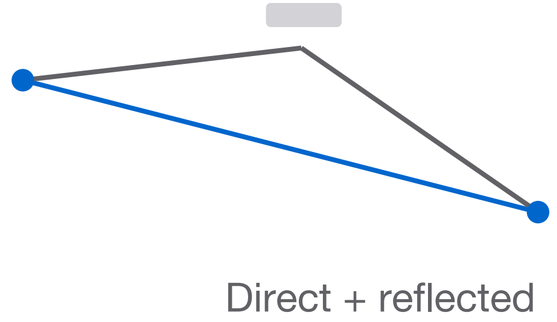
Navigation claim misled



A clear voice, wrong directions.

Multipath

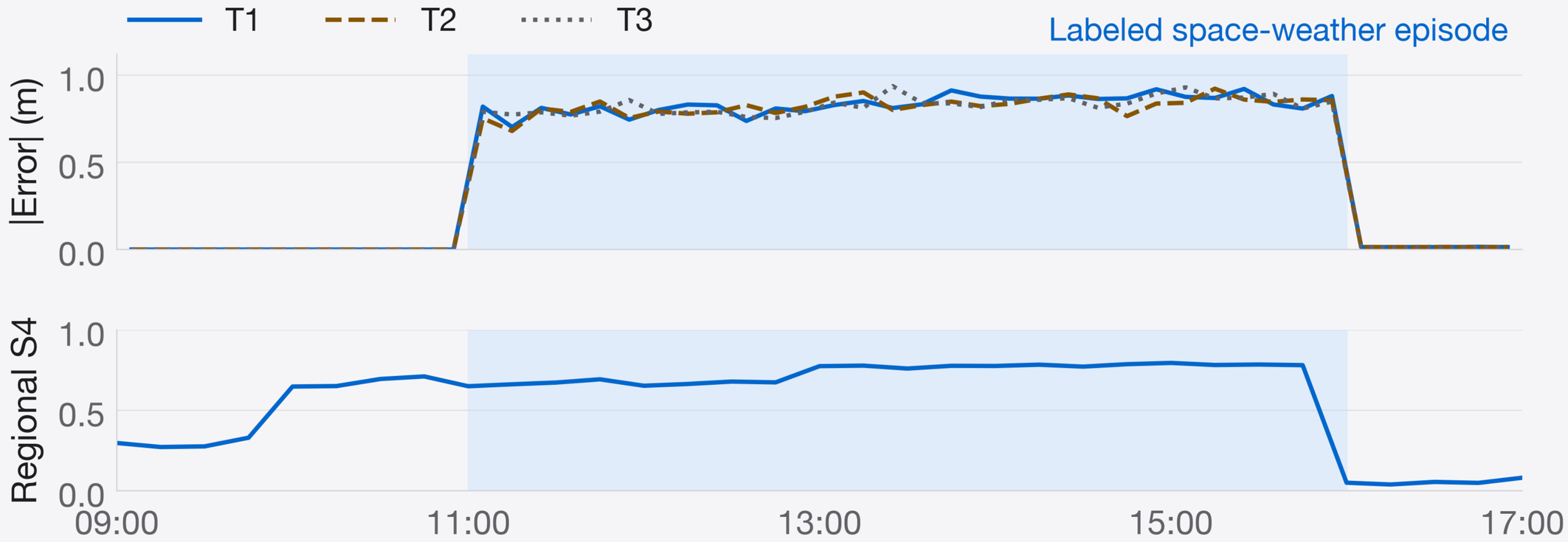
Signal takes extra paths



An echo arrives by a longer route.

Signal health and position truth are different questions.

Put the evidence on the same clock.



Public development data · May 5, 2026 (UTC) · Ten-minute medians can hide brief events

The same drift.
Five different next moves.

Space weather

Halt or degraded mode

Jamming

Halt and report

Spoofing

Halt, cross-check, report

Equipment fault

Service machine; others continue

Benign multipath

Continue; dismiss false alarm

Causes can overlap. Distinct causes need distinct detections.

A useful call.
A reason
to trust it.

Action.
Evidence.
Honest uncertainty.

T2 · Guidance degraded

Recommended action

Service T2.
Other tractors continue.

Evidence to check

Correction link unavailable on T2.
Neighboring tractors remain stable.

Still uncertain

Additional receiver checks needed.

Illustrative equipment-fault case

Learn from what’s known. Investigate what’s new.

12

Labeled development episodes

48 simulated hours

8

Unlabeled evaluation

24 simulated hours

3 tractors · 72 simulated hours · 1 Hz · Farm + SDA feeds · Synthetic and unclassified

Build. Explain. Defend.

Friday 4:00 pm to Saturday 4:00 pm CT · Individual work

24 hours to build

5 min demo

3 min Q&A

One pull request by Saturday 4:00 pm CT

GitHub username only · Repository public when you submit · Any language and AI assistants

Step-by-step: bluestaq.github.io/ndsu-hackathon-submissions

Show the quality of your judgment.

Working software.

Useful decisions.

Honest limits.

Technical execution	**30**
Mission + decision quality	**25**
Innovation + research	**20**
Assurance + rigor	**15**
Demo + communication	**10**

The diagnostic supports judging. It adds no event points.

Build your future.

ONE WINNER · BOTH AWARDS

Paid summer internship

Bluestaq HQ
Colorado Springs

Full one-year scholarship

Tuition paid for one year.
Continue your education.

Internship subject to standard eligibility and hiring requirements.

BLUESTAQ™ | NDSU NORTH DAKOTA STATE UNIVERSITY

Make your skills matter.



Build for the person
who has to decide.

Open the guide.
Run the baseline.
Make your first improvement.

bluestaq.github.io/ndsu-hackathon-submissions

One winner · Paid summer internship + full one-year scholarship, tuition paid

The event, at a glance.

Eligibility	NDSU undergraduate students who are U.S. citizens
Timing	Build Fri 4:00 pm – Sat 4:00 pm CT; demos and judging follow
Work	Individual; no competitor collaboration or solution sharing
Tools	Any language, internet resources and AI assistants
Submit	One pull request by Sat 4:00 pm CT: submissions/<github-username>/submission.json
Present	Five-minute working demo + three-minute Q&A
Award	One winner: paid summer internship + full one-year scholarship, tuition paid

Internship subject to standard eligibility and hiring requirements. universityprograms@bluestaq.com

Cause and action.

SPACE_WEATHER	HALT_OR_DEGRADED_MODE
JAMMING	HALT_AND_REPORT
SPOOFING	HALT_CROSSCHECK_AND_REPORT
EQUIPMENT_FAULT	SERVICE_MACHINE_OTHERS_CONTINUE
MULTIPATH	CONTINUE_FALSE_ALARM

Use separate detections for distinct concurrent causes.

Read the tractor in three ways.

Motion

Position

Where the receiver claims to be.

Speed

Working or parked?

Reception

C/N0

Signal clarity relative to noise.

Satellites

Number used in the solution.

Guidance

Cross-track error

Deviation from the intended row.

Fix + base link

Solution mode and corrections.

One record = one tractor at one UTC second.

RTK_FIXED is a solution claim. Reported coordinates are not trusted ground truth.

Context has a different clock.

Kp

Broad geomagnetic disturbance · Scale 0–9

Every 3 hours

Dst

Storm-time magnetic change · Nanoteslas

Every hour

S4

Signal-amplitude flicker
Regional proxy

Every 15 minutes

Advisories add notices. RFI reports can be delayed, missing, or false.

Conceptual sampling density · SDA = Space Domain Awareness · All challenge feeds are simulated

Make the first improvement real.

1

Run it.

Get the baseline working.

2

Question it.

Inspect one labeled case.

3

Improve it.

Measure one useful change.

AI assistants are allowed. Understand and defend what you submit.

Your first working result.

Python 3.11 or newer · From the student-package root

```
python -m pip install -r requirements.txt
python submission/baseline.py --data data --output submission.json \
--github-handle YOUR_GITHUB_USERNAME --repo-url https://github.com/YOU/YOUR_REPO
python submission/validate_submission.py submission.json
```

Each detection: times · tractors · cause · action · confidence · evidence

Evaluation: May 6, 2026 00:00 UTC to May 7, 2026 00:00 UTC

GitHub username only: never include your email or student ID · Validation checks format

The diagnostic is evidence.

25 × macro-F1 + 15 × mean onset credit

Matching	One-to-one; temporal IoU ≥ 0.30; tractor-set IoU ≥ 0.50.
Onset	Full credit within 5 minutes; declines to zero at 60 minutes.
False alarms	Extra detections reduce precision; multipath exemption has specific limits.
Interpretation	Retrospective onset accuracy is not measured online alert latency.

Read page 4 for matching priorities, duplicate rules and the full multipath exemption.

Why it matters: sources (1 of 2).

FAO · Rome Declaration, World Food Summit · Nov. 13, 1996
https://www.fao.org/4/w3613e/w3613e00.htm

FAO · Food Security policy brief, Issue 2 · June 2006
https://www.fao.org/fileadmin/templates/faoitaly/documents/pdf/pdf_Food_Security_Cocept_Note.pdf

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https://esmis.nal.usda.gov/sites/default/release-files/jq085j963/js956k123/3r074z43v/ERSF-04-11-1973_Communist_Areas.pdf

GAO · The Russian Wheat Sales, testimony · Mar. 8, 1973
https://www.gao.gov/assets/094557.pdf

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https://ntrs.nasa.gov/citations/19790010237

NASA · The LACIE Symposium, plenary proceedings · Oct. 1978
https://ntrs.nasa.gov/citations/19790006278

USDA ERS · Why Have Food Commodity Prices Risen Again? WRS-1103 · June 2011
https://ers.usda.gov/sites/default/files/_laserfiche/outlooks/40481/7392_wrs1103.pdf

United Nations · Black Sea Grain Initiative
https://www.un.org/en/black-sea-grain-initiative

NASA Harvest · Tracking Ukraine’s winter crops during the war · Oct. 28, 2026
https://www.nasaharvest.org/news/how-nasa-harvest-is-tracking-ukraines-winter-crops-during-the-war

White House · HSPD-9, Defense of U.S. Agriculture and Food · Jan. 30, 2004
https://georgewbush-whitehouse.archives.gov/news/releases/2004/02/20040203-2.html

Why it matters: sources (2 of 2).

White House · NSM-22, Critical Infrastructure Security and Resilience · Apr. 30, 2024
https://bidenwhitehouse.archives.gov/briefing-room/presidential-actions/2024/04/30/national-security-memorandum-on-critical-infrastructure-security-and-resilience/

CISA · Food and Agriculture Sector
https://www.cisa.gov/topics/critical-infrastructure-security-and-resilience/critical-infrastructure-sectors/food-and-agriculture-sector

Federal Register · Executive Order 13905, PNT resilience · Feb. 12, 2020
https://www.federalregister.gov/documents/2020/02/18/2020-03337/strengthening-national-resilience-through-responsible-use-of-positioning-navigation-and-timing

USDA · Farm Security is National Security, press release · July 8, 2025
https://www.usda.gov/about-usda/news/press-releases/2025/07/08/farm-security-national-security-trump-administration-takes-bold-action-elevate-american-agriculture

USDA NASS · North Dakota Agricultural Statistics 2025 (2024 data) · Aug. 2025
https://www.nass.usda.gov/sites/default/files/Statistics_by_State/North_Dakota/Publications/Annual_Statistical_Bulletin/2025/ND-Annual-Bulletin25.pdf

USDA ERS · Precision Agriculture in the Digital Era, EIB-248 · Feb. 2023
https://ers.usda.gov/sites/default/files/_laserfiche/publications/105894/EIB-248.pdf

RTI International for NIST · Economic Benefits of GPS · June 2019
https://www.rti.org/publication/economic-benefits-global-positioning-system-gps/fulltext.pdf

Traficom · Disturbances in satellite navigation in Finland, 2024 · May 10, 2024
https://www.traficom.fi/en/news/situational-picture-disturbances-satellite-navigation-finland-2024

How space feeds a nation: sources.

NASA · SMAP (Soil Moisture Active Passive) mission
https://science.nasa.gov/mission/smap/

NOAA NESDIS · GOES
https://www.nesdis.noaa.gov/geostationary-operational-environmental-satellites-goes

NOAA NESDIS · Joint Polar Satellite System
https://www.nesdis.noaa.gov/our-satellites/currently-flying/joint-polar-satellite-system

NOAA NESDIS · SWFO-L1, renamed SOLAR-1, reaches L1 · Jan. 27, 2026
https://www.nesdis.noaa.gov/news/swfo-l1-renamed-solar-1-reaches-final-destination-one-million-miles-earth

NOAA SWPC · SOLAR-1 is now operational · June 10, 2026
https://www.spaceweather.gov/news/solar-1-now-operational-and-data-available

NOAA NESDIS · DSCOVR and 15–60 minute solar-wind warnings
https://www.nesdis.noaa.gov/news/dscovr-poised-liftoff-extending-critical-space-weather-alerts-forecasts

NASA · CYGNSS mission
https://science.nasa.gov/mission/cygnss/

NASA PO.DAAC · CYGNSS Level 3 soil moisture, v3.2
https://podaac.jpl.nasa.gov/dataset/CYGNSS_L3_SOIL_MOISTURE_V3.2

HawkEye 360 · GNSS interference detection · July 29, 2025
https://www.he360.com/hawkeye-360-launches-advanced-gnss-interference-detection-capabilities-to-enhance-defense-and-intelligence-operations/

GPSJam · FAQ: ADS-B-derived interference maps
https://gpsjam.org/faq

USDA FAS · World Agricultural Production, WAP 10-26 · Oct. 2026
https://apps.fas.usda.gov/psdonline/circulars/production.pdf

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https://www.bluestaq.com/us/site/about.html

Bluestaq · /ARQ for agriculture · updated Sept. 9, 2026
https://www.bluestaq.com/ask-assets/content/vert-agriculture.html

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https://spacenews.com/bluestaq-launches-bluestaq-arq-defense-grade-data-infrastructure-now-available-for-commercial-government-and-enterprise-organizations/

Bluestaq release via SpaceNews · primary developer of the UDL · Apr. 8, 2025
https://spacenews.com/bluestaq-expands-globally-with-50m-from-one-bow-river-launches-uk-operations/

Space Systems Command · UDL provides data solutions at the speed of battle · Mar. 11, 2024
https://www.ssc.spaceforce.mil/Newsroom/Article/4039108/space-systems-commands-udl-provides-data-solutions-at-the-speed-of-battle

Breaking Defense · Calvelli report to Congress on the UDL · June 27, 2024
https://breakingdefense.com/2024/06/calvelli-details-plans-to-better-integrate-unified-data-library-into-space-force-ops/

Bluestaq · UDL Python SDK, public API reference (Apache-2.0)
https://github.com/Bluestaq/udl-python-sdk/blob/main/api.md

GPS.gov · GPS space segment, about 20,200 km
https://www.gps.gov/space-segment

NOAA NESDIS · GOES, about 35,800 km
https://www.nesdis.noaa.gov/geostationary-operational-environmental-satellites-goes

Grounded in the guide.

Official event guide, v1.3

../StraightRows_Student_Package/docs/Bluestaq_NDSU_Straight_Rows_Hackathon_Official.pdf

Student package: README, notebook, baseline and validator

../StraightRows_Student_Package/README.md

Bluestaq: data infrastructure and Space Domain Awareness

https://www.bluestaq.com/us/

NOAA: G5 conditions observed · May 10, 2024

https://www.swpc.noaa.gov/news/g5-conditions-observed

404 Media: tractor GPS disruption · May 12, 2024

https://www.404media.co/solar-storm-knocks-out-tractor-gps-systems-during-peak-planting-season/

Future Farming: reported interference at Finnish farms · June 14, 2024

https://www.futurefarming.com/tech-in-focus/autonomous-semi-autosteering-systems/tractors-scrambled-on-the-eastern-border-of-finland-by-russian-interference/

Understand the signals.

ESA Navipedia · GNSS measurements and timing

https://gssc.esa.int/navipedia/index.php/GNSS_Basic_Observables

ESA Navipedia · Real-Time Kinematic positioning

https://gssc.esa.int/navipedia/index.php/Real_Time_Kinematics

NOAA · Space weather and GPS systems

https://www.spaceweather.gov/impacts/space-weather-and-gps-systems

NOAA · Ionospheric scintillation and S4

https://www.spaceweather.gov/phenomena/ionospheric-scintillation

NOAA · Planetary Kp index

https://www.spaceweather.gov/products/planetary-k-index

World Data Center, Kyoto · Dst index

https://wdc.kugi.kyoto-u.ac.jp/dstdir/dst2/onDstindex.html

Design & image credits.

Apple Human Interface Guidelines · Design principles
https://developer.apple.com/design/human-interface-guidelines/design-principles

Apple HIG · Layout, typography, color and accessibility
https://developer.apple.com/design/human-interface-guidelines/

Apple HIG · Motion
https://developer.apple.com/design/human-interface-guidelines/motion

Apple Gazette · Storytelling and branding
https://www.applegazette.com/resources/apples-marketing-lessons-in-storytelling-and-branding/

Planting: Wikideas1 · CC0 · Wisconsin (representative scene)

Aurora: NASA / Bill Dunford · Public domain · May 11, 2024

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North Dakota from orbit: NASA JSC / ISS Expedition 37 · Public domain · Sept. 15, 2013

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